system-enforced requirement that they be changed on the first sign-in. But for other kinds of credentials like digital certificates, that isn't usually necessary.

Regardless of type, the result of enrollment is a new relationship anchored by the identifier, protected by authentication, and containing the attributes and other information necessary for the relationship to function.

3.4.2 Identity proofing and assurance

The kind of enrollment you use depends on the requirements of the digital relationship you're creating. One of the most important aspects is called the *level of assurance* and is determined by how strongly you need to be sure of the identity of the entity that you're enrolling. The process of verifying an entity during enrollment is called *proofing*.

NIST Special Publication 800-63-3 defines three levels of assurance (LOAs) based on the rigor of the proofing process. Table 3.1 summarizes the NIST identity assurance levels.

Table 3.1 Summary of NIST Identity Assurance Levels (IAL), which define the strength of identity proofing during enrollment based on the necessary confidence in the claimed identity.

IAL Level	Description	Enrollment Process Example
IAL1 (low)	Self-asserted identity; no formal identity proofing.	Signing up for an email account or a social media profile.
IAL2 (moderate)	Identity verified using trusted documents (e.g., driver's license); can be remote.	Opening an online bank account with ID and selfie verification.
IAL3 (high)	In-person verification with biometrics or cryptographic binding to the identity.	Getting a passport, security clearance, or high-assurance enterprise login.

Which of these three levels you choose depends on the amount of risk your system can tolerate. This might be something you decide for yourself but is often driven by regulation. For example, anti-money-laundering regulations mandate a *Know Your Customer (KYC)* process that translates, in most cases, to an IAL2 requirement. Consequently, a financial services institution will